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Gravity-Based Community Vulnerability Evaluation Model in Social Networks: GBCVE

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***Abstract*—The usage of social media around the world is ever-increasing. Social media statistics from 2019 show that there are 3.5 billion social media users worldwide. However, the existence of community structure renders the network vulnerable to attacks and large-scale losses. How does one comprehensively consider the multiple information sources and effectively evaluate the vulnerability of the community? To answer this question, we design a gravity-based community vulnerability evaluation (GBCVE) model for multiple information considerations. Specifically, we construct the community network by the Jensen–Shannon divergence and log-sigmoid transition function to show the relationship between communities. The number of edges inside community and outside of each community, as well as the gravity index are the three important factors used in this model for evaluating the community vulnerability. These three factors correspond to the interior information of the community, small-scale interaction relationship, and large-scale interaction relationship, respectively. A fuzzy ranking algorithm is then used to describe the vulnerability relationship between different communities, and the sensitivity of different weighting parameters is then analyzed by Sobol' indices. We validate and demonstrate the applicability of our proposed community vulnerability evaluation method via three real-world complex network test examples. Our proposed model can be applied to find vulnerable components in a network to mitigate the influence of public opinions or natural disasters in real time. The community vulnerability evaluation results from our proposed model are expected to shed light on other properties of communities within social networks and have real-world applications across network science.**

***Index Terms*—Community vulnerability, complex networks, gravity model, Jensen–Shannon divergence, log-sigmoid transition, science of cities, social networks.**

I. Introduction

SOCIAL media has become an indispensable aspect of modern daily life and changed the way how people interact with one another [1], [2]. Sharing news on Twitter and chatting with friends via Telegram and WhatsApp has become the norm, rather than the exception. To give another example, recent waves of controversies surrounding genetic engineering have spilled into popular science in Twitter battles between reputable scientists and their followers as well [3]. In order to assess people's influence in social media, social networks have routinely been proposed to describe the relationships between individuals [4]–[7]. In social networks, each individual has a community characterized by close internal connections and sparse external connections structure. Such structure is also reflected in many real-world networks and plays an important role in the study of network science [8], [9]. The study of their structure has numerous real-world applications [10]: community structure in social networks can be analyzed to determine users with strong influencing ability [11]; community reliability in the power network is analyzed to maintain normal operation [12]. Finally, the community connectivity of the road network can be measured to reduce any disruption to traffic flow [13]. Due to the ubiquity of community structure in many disciplines, it has become a popular topic in the field of network science [14], and many methodologies [15]–[18] have been proposed to solve problems related to the community of social networks. These methodologies include game theory [19]–[23], opinion dynamics [24]–[26], synchronization analysis [27]–[29], and multiobjective optimization [30]–[33].

In general, research pertaining to network community attempts to answer two fundamental questions. The first question asks: how does one detect the community structure in a network effectively and accurately. Detecting community structure is of great importance in sociology, biology, computer science, and disciplines where systems are often represented as graphs [34]–[36]. Numerous algorithms [37]–[41] have been proposed to address this problem arising from different types of networks. Newman [42] developed a split hierarchical clustering method to detect the community structure of large-scale network effectively, and Fortunato [34] applied an interdisciplinary method to detect community structure from several real-world networks. The other question asks how can one quantitatively describe the common properties in each community. It is worth noting that these community properties may shed light on the other network properties. For example, the similarity between each pair of nodes can be measured to find the same group across multilayer social network [43]–[46]. The subset of highly influential individuals is identified to reduce the scale of disaster and rumor transmission [47]–[51]. The resilience of community can be quantified

Manuscript received May 7, 2021; revised August 23, 2021; accepted October 12, 2021. This work was supported in part by the Ministry of Education of Singapore Academic Research Fund Tier 2 under Grant MOE-T2EP50120-0021, and in part by the Singapore University of Technology and Design under Grant SRG SCI 2019 142. This article was recommended by Associate Editor S. Ozawa. *(Corresponding author: Kang Hao Cheong.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCYB.2021.3123081.

Digital Object Identifier 10.1109/TCYB.2021.3123081

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to describe how the initial set of community structures can recover after the disruption [52]–[54].

There has been immense interest in the vulnerability evaluation of community [55] because this property can help to unravel many critical components of social networks. Resources of the network can be prioritized to reduce any potential damage to specific vital nodes and edges. Dispatch and assistance can then be used to improve the capability in emergency handling during sudden attacks, and the performance of each component in the network can still be well protected [56]. Irrefutably, people do not need to wait until the urban road network encounters congestion, public opinions begin to spread on social networks, or an epidemic that spreads across the world, before they start to pay attention to the vulnerability of each component in networks. Therefore, researchers need to evaluate the vulnerability of communities in the network according to the network structure, and make adjustments accordingly to improve the safety performance in advance. All these real-world situations that we have mentioned highlight the importance of studying the vulnerability of the community as it is necessary to identify these critical communities in order to enhance the security of networks. The vulnerability of community is the susceptibility of communities to negative impacts from hazardous events [57] and how the topological structure of community is affected by the removal of a finite number of edges and nodes [58], which may lead to the overload of components and cascading failure, such as the user (node) attacked by hackers in the social network, the congested road (edge) in a traffic network, and the transmission line (edge) failures in a power grid network. Although the definition of community in the network is a substructure with close internal connections and sparse external connections [59], in practice, when the community is attacked (node or edge removal), both internal resource allocation and external resource assistance are required to protect the structure of the community represented by the internal and external connections. Thus, the connectivity within the community, as well as with external communities should be considered when analyzing the vulnerability of community.

The prior research in network science has primarily focused on evaluating community vulnerability with the corresponding protection methods [58], [60]–[62]. Recently, Rocco and Ramirez-Marquez [63] have developed a qualitative metric to evaluate community vulnerability by focusing on the connectivity between every two connected communities. This qualitative metric provides some insights into the relative strength of each community, and has been used in several real-world complex networks. The important point to note here is that connectivity has now been shown to be an important consideration in individual's resistance to disasters in the network, that is, the vulnerability of community. Based on this qualitative metric, Wei *et al.* [64] then proposed a generalized model by considering both the outer connectivity of a community and its inner structure, but limited in its application when tested on the real-world network. However, some of these works have failed to evaluate community vulnerability in certain realistic structure connectivity.

In order to fill this important gap, we propose a comprehensive evaluation method pertaining to community vulnerability in social networks called the gravity-based community vulnerability evaluation (GBCVE) model. Specifically, the community structure is detected by Newman's modularity [59] from social networks. The basic information can be obtained from the structure directly, containing the number of edges inside the community (EIC) and the number of edges outside the community (EOC). The distance, which describes the difference between every two communities, is then obtained by the Jensen–Shannon divergence [65] and log-sigmoid transition technique. The community network, which is a fully connected undirected weighted network, can be constructed by the computed distance. Furthermore, the gravity index (GI) of each community is then calculated based on the gravity model from the new community network. Our proposed model takes into account these three factors with weighting parameters. These three factors correspond to the different aspects of information under consideration: 1) the interior information of each community; 2) the small-scale interaction relationship; and 3) large-scale interaction relationship, respectively. Under the special weighting parameter setting, our method is equivalent to the classical community vulnerability evaluation method [63]. Finally, the vulnerability of each community can be calculated by our proposed model, with a fuzzy ranking algorithm to determine the vulnerability order. This fuzzy ranking algorithm can be used to describe the vulnerability relationship between different communities. It is reasonable to use such an algorithm because different aspects of community information can then be considered in a holistic manner. In addition, the sensitivity of these weighting parameters is analyzed by Sobol' indices.

Our approach has been experimentally evaluated by three real-world complex network test examples. These experimental results indicate the reasonableness and effectiveness of our proposed method. Our proposed method can consider multiple information sources, thereby effectively distinguishing the vulnerabilities of communities in all test examples, whereas the classical method has not been able to do so. Importantly, the vulnerability order recognized by our approach corresponds to the actual community structure and the connectivity in their entirely. The quantitative analysis of communities also supports our approach. The main contributions of this article are summarized as follows.

1) A new and comprehensive evaluation model is proposed to measure the vulnerability of communities in the network, and three kinds of connectivity factors with weighting parameters are considered. This model can degenerate to the classical method with special weighting parameters.
2) A gravity-based model is applied to obtain the large-scale interaction relationship between communities, which takes into account both the information of the community itself and the relationship between the communities. In addition, the distance between communities is determined by their similarity rather than their physical distance because the physical distance only reflects

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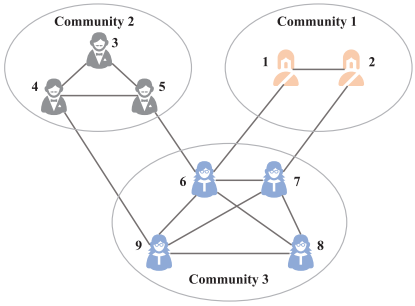

Fig. 1. Example social network with nine nodes.

their distance, but not the relationship (which is the focus of vulnerability).

3) The fuzzy ranking method is applied to determine the vulnerability order, which can better describe the relationship between communities by their gap rather than only giving a vulnerability ranking.
4) The performance of our proposed method is evaluated by three different real-world complex networks. The experimental results show that our proposed method is superior to the classical algorithm, and can even identify communities that are not distinguishable by the classical algorithm.

The remainder of this article is organized as follows. The problem statement is given in Section II. Section III develops this community vulnerability evaluation model in detail, and is further divided into four sections. The experimental evaluation involving three real-world complex network test examples is given in Section IV, and the discussion on the effectiveness and reasonableness of our proposed model is also given. Finally, conclusions and future work are discussed in Section V.

II. Problem Definition

For a given social network $G(N, E)$, it comprises multiple social individuals and relationships (links) between individuals. Each individual in the network has different connectivity strengths, so the network can be divided into several communities. One example of the social network with nine individuals and 14 links is shown in Fig. 1, and this social network can be divided into three communities (c_1, c_2, and c_3) by Newman's modularity method [59]. We aim to evaluate the vulnerability of communities by considering multiple structure properties of each community, and then obtain the vulnerability order in social networks to guide the establishment of social networks.

Our problem statement is defined as follows.

Input: The input information is the structure of the social network $G(N, E)$, where N is the set of individuals and E is the set of relationship between individuals. The topological relationship of the social network is given by the adjacency matrix A, and the element a_{ij} in A indicates whether there is a connection between individual i and individual j.

Output: When the social network is divided into several communities, our proposed method can evaluate the vulnerability of community c_i and output the vulnerability fuzzy ranking in the network.

III. Methodology

In the real-world, every individual has their own community, and the vulnerability of communities also affects individuals' activities. Thus, a novel method, called the GBCVE model, is proposed in this article to consider: 1) the property of community itself (interior information); 2) the relationship between the chosen community and directly connected neighboring communities (small-scale interaction relationship); and 3) the relationship between the chosen community with all other communities, regardless of their connection and distance (large-scale interaction relationship). By considering community information described above, coupled by the weights given to the information in different situations, makes GBCVE more reasonable and effective than existing methods to determine vulnerability in community.

For any given social network $G(N, E)$, the vulnerability of communities can be obtained by the following steps.

1) *Divide the Network Into Several Communities:* The entire network G can be divided into several nonintersecting communities, and each individual belongs to only one community c_i.
2) *Obtain the Basic Information of Each Community:* With the obtained community structure, basic information of the community can then be retrieved, containing the number of EIC and the number of EOC.
3) *Construct Community Network via the Difference Between Communities:* According to the structure of communities, the distance describing the difference between communities can be obtained by the Jensen–Shannon divergence and log-sigmoid transition function. The community network, a fully connected undirected weighted network, can then be constructed by the computed distance.
4) *Construct the Large-Scale Relationship Between Different Communities:* Given the computed distance between communities, the large-scale relationship (gravity index and GI) between each pair of communities can be determined by the gravity model.
5) *Compute the Vulnerability of Each Community:* With the three obtained factors of each community (EIC, EOC, and GI), the vulnerability of community can then be calculated based on our comprehensive model.
6) *Obtain the Fuzzy Ranking of Community Vulnerability:* After the vulnerability of community calculation process, the fuzzy ranking order of community vulnerability can be obtained.

The framework of this proposed GBCVE model is shown in Fig. 2. In order to introduce the model in details, the rest of this section is divided into four sections. Specifically, Section III-A introduces how to obtain the community structure from social network, which corresponds to step 1. Section III-B obtains the basic information of each community, which is step 2. The large-scale relationship in community can be measured by divergence, transition technique, and gravity model as given in Section III-C, which contains steps 3 and 4. Finally, the vulnerability evaluating and ranking

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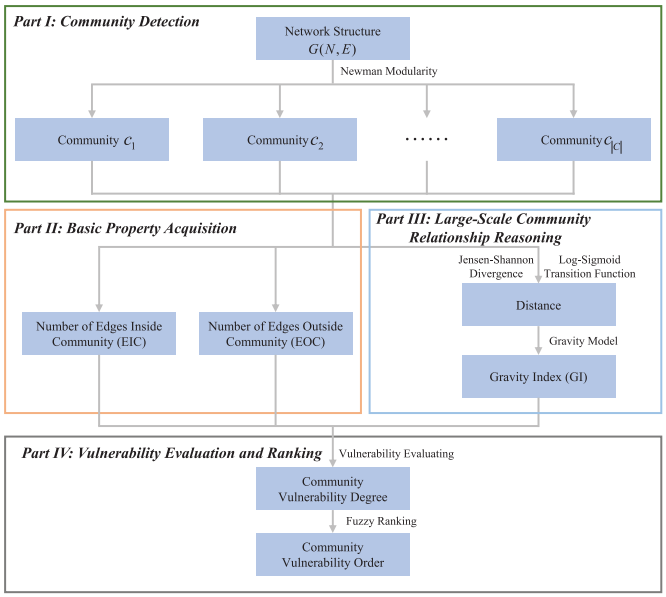

Fig. 2. Framework of this proposed GBCVE model.

process are given in Section III-D, corresponding to step 5 and 6.

A. Part I: Communities Detection

A social network can be denoted as $G(N, E)$, where $N = \{1, 2, \ldots, |N|\}$ and $E = \{1, 2, \ldots, |E|\}$ are the set of nodes and edges, respectively, $|N|$ and $|E|$ are the whole number of nodes and edges, respectively. The topological relationship in the network is represented by the adjacency matrix A, whose size is $|N| \times |N|$. The element a_{ij} in A determines whether there is an edge between node i and node j, that is, $a_{ij} = 1$ means the existence of edges and *vice versa.*

Definition 1: The modularity of network with $|C|$ communities is denoted as Q [59], and defined as follows:

$$Q = \sum_{c_k=1}^{|C|} \left(\frac{|E_{c_k}|}{|E|} - \left(\frac{\sum_{i \in N_{c_k}} d_i}{2|E|} \right)^2 \right) \tag{1}$$

where c_k is the chosen community, $|C|$ and $|E|$ are the total number of communities and edges in the network, respectively, $|E_{c_k}|$ is the number of edges in the community c_k, N_{c_k} is the set of nodes in community c_k, and d_i is the degree of node i. The degree of node refers to the number of edges connected with the chosen node.

The value of Q is a good indicator of the strength of the community structure.

1) $Q = 0$ represents all nodes (i.e., $|N|$ nodes) are divided into one single community, and there is no community structure in the social network.
2) $Q \in (0, 1)$ means the existence of community structure, and different values of Q indicate the strength of the community structure.
3) $Q = 1$ indicates the existing community structure is very strong.

In addition, Newman and Girvan [59] have given the suggestion that the value of Q between 0.2 and 0.7 indicates a strong community structure. The community structure detection algorithm is to find the maximum value of Q, and the detailed steps are given as follows.

Step 1: Treat every node in the network as a separate community, that is, there is $|N|$ communities at the beginning, and obtain Q_0 in this situation. Any two communities will be merged to find the maximum value of Q in the following steps.

Step 2: Combine any two communities into a community, and calculate $Q_t(i, j)$ of the integrated community structure (combined community i and j in the tth integration).

Step 3: Obtain the modularity change $\Delta Q_t(i, j) = Q_t(i, j) - Q_{t-1}$ after the integration of the two communities.

Step 4: Find the maximum value of ΔQ_t in the tth integration, that is, $\Delta Q_t = \max_{i,j} \Delta Q_t(i, j)$, and this structure with modularity Q_t is chosen as the integration strategy.

Step 5: Repeat steps 2–4 until $\Delta Q_t < 0$.

B. Part II: Basic Property Acquisition

With the obtained community structure, the basic information of each community can be retrieved to describe their properties. The number of EIC and the number of EOC are two important properties to describe community vulnerability.

Definition 2: The EIC for community c_i is obtained as follows:

$$\eta_i = \sum_{j,k \in c_i} a_{jk} \tag{2}$$

where a_{jk} represents whether there is an edge between node j and node k, and node j and node k are completely inside the community. Therefore, EIC demonstrates the interior connectivity of the community.

Definition 3: The EOC for community c_i is obtained as follows:

$$\sigma_i = \sum_{j \in c_i, k \notin c_i} a_{jk}. \tag{3}$$

Unlike EIC, EOC describes the accessibility between the chosen community and other communities (node j belongs to community c_i, but node k belongs to other communities). Large σ_i means this community c_i is convenient to communicate with other communities, thereby demonstrating the small-scale interaction relationship.

C. Part III: Large-Scale Community Relationship Reasoning

The large-scale relationship, together with the vulnerability evaluation factor, is then obtained from this constructed community network.

1) Community Difference: The distance between communities can be regarded as the physical distance, that is, the sum of edge lengths. However, this only reflects the distance but not the relationship between communities, which is the focus

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of vulnerability consideration. Therefore, community properties are now considered to calculate the distance between communities.

Assume that the community with the largest number of nodes has κ nodes, that is, $\kappa = \max_{c_k} |N_{c_k}|$. There is $|N_{c_i}|$ nodes in community c_i, and the probability set of community c_i is given as follows:

$$P_i = \left[p_i(1), p_i(2), \ldots, p_i(m), \ldots, p_i(\kappa)\right]. \tag{4}$$

The setting of κ renders all probability sets to have the same size. $|N_{c_i}|$ is usually less than κ, and equals κ when c_i is the largest community. Therefore, some elements of P_i will be equal to zero when $|N_{c_i}| < \kappa$, and the element of P_i is defined as follows:

$$p_i(t) = \begin{cases} \frac{d_t}{\sum_{t \in N_{c_i}} d_t}, & t \le \left|N_{c_i}\right| \\ 0, & t > \left|N_{c_i}\right| \end{cases} \tag{5}$$

where d_t is the degree of node t. Each node in community has one probability value, and the rest of the elements will be equal to zero to make P to have the same size. To avoid the sequence of the elements affecting the communities' difference measurement, the probability set P_i is sorted in descending order

$$P_i' = \left[p_i'(1), p_i'(2), \ldots, p_i'(t), \ldots, p_i'(\kappa)\right]. \tag{6}$$

Definition 4: The Jensen–Shannon divergence is widely used to measure the difference between probability distribution in information theory, so we use it here to measure the difference between two community structure μ_{ij}

$$\begin{aligned} \mu_{ij} &= D_{JS}\left(P_i' || P_j'\right) \\ &= \frac{1}{2} \times \left(\sum_{t=1}^{\kappa'} p_i'(t) \ln \frac{p_i'(t)}{p_{\text{con}}'(t)} + \sum_{t=1}^{\kappa'} p_j'(t) \ln \frac{p_j'(t)}{p_{\text{con}}'(t)} \right) \end{aligned} \tag{7}$$

where $P_{\text{con}}' = ([P_i' + P_j']/2)$, P_i' and P_j' are two descending order of probability sets for c_i and c_j. κ' is the smaller one between $|N_{c_i}|$ and $|N_{c_j}|$, that is, $\kappa' = \min\{|N_{c_i}|, |N_{c_j}|\}$, which can avoid $p_i'(t)/p_{\text{con}}'(t)$ or $p_j'(t)/p_{\text{con}}'(t)$ being 0 or infinity.

The Jensen–Shannon divergence between two communities is symmetrical, and it falls in [0, 1]. This means the larger the value, the greater the difference between the two communities. The log-sigmoid transition function (S-type growth curve) is then used to transform the Jensen–Shannon divergence into the distance between each pair of communities, changing the original degree of distinction ($\upsilon_{ij} - \upsilon_{jk} \neq \mu_{ij} - \mu_{jk}$).

Definition 5: The distance υ_{ij} obtained by log-sigmoid transition function is defined as follows:

$$\upsilon_{ij} = \frac{1}{1 + e^{-\varphi \cdot \mu_{ij}}}, i \neq j \tag{8}$$

where φ is the fitting parameter to adjust the performance of this model. $i \neq j$ means there is no self-loop. With the change of φ, the distance between communities will also change. The fitting parameter φ is chosen as 3 in this model to achieve a better effect (shown in Fig. 3).

The obtained distance will form a symmetrical community distance matrix and can be used to constructed a fully connected undirected weighted community network.

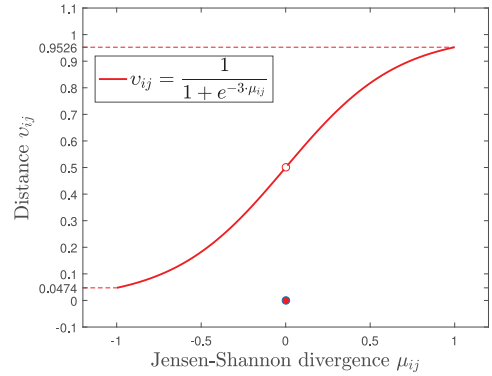

Fig. 3. Log-sigmoid transfer function expression with $\varphi = 3$.

2) Large-Scale Community Relationship: Based on the distance between each pair of communities, we now use the gravity model algorithm [66] to obtain the large-scale community relationship. The distance in the network represents the similarity between every two communities rather than the physical distance, because the physical distance cannot reflect the relationship between communities, which is the crux of vulnerability evaluation. In the real world, more individuals and components in a group (community) represent more resources, information flow, materials, and power in the community, thus leading to a more reliable community. In addition, there will be more interaction between two communities when the distance is shorter (more similar in this model). The interaction means the mutual assistance between communities, thereby increasing the ability to handle emergency situations and reducing the vulnerability under any accident. Overall, closer distance and more individuals in communities represent lower vulnerability. The influence of the number of nodes and the distance on the vulnerability is similar to the definition of the gravity model. In addition, the effect of the gravity will be weaker when the objects are further away, but it has an infinite range without the limitation of distance. Therefore, the gravity-based vulnerability model, which considers multiple information, is used to represent the large-scale interaction relationship to measure the community's ability to resist negative impacts from events. The community with higher GI in this model means that they have more communities with rich resources that are close to them, thus this particular community is said to be less vulnerable and have the ability to resist negative impacts from events.

Definition 6: Generally speaking, the community with more nodes will have lower vulnerability. Thus, the GI of each community is obtained by the gravity model

$$\gamma_i = \sum_{c_j \in C, c_j \neq c_i} \frac{\left|N_{c_i}\right| \times \left|N_{c_j}\right|}{\upsilon_{ij}^2} \tag{9}$$

where $|N_{c_i}|$ and $|N_{c_j}|$ are the number of nodes in community c_i and c_j, respectively, υ_{ij} is the distance between c_i and c_j, and γ_i is the sum of all other communities' impact on c_i. The closer the two communities with lower vulnerabilities will make the two groups more reliable.

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TABLE I
VULNERABILITY OF COMMUNITIES IN THE SOCIAL NETWORK EXAMPLE SHOWN IN FIG. 1

Community c_i	η_i	σ_i	γ_i	ζ_i	ξ_i	V_i [63]	R_i [63]
c_1	0.1667	0.5000	0.6814	17.6099	17.6099	2	2
c_2	0.5000	0.5000	0.9217	4.3133	4.3133	2	2
c_3	1	1	1	1	1	1	1

D. Part IV: Vulnerability Evaluating and Ranking

1) Community Vulnerability Evaluating: With the obtained EIC, EOC, and GI, the vulnerability of communities can then be evaluated and ranked to analyze their properties. EIC considers the interior information; EOC considers the interaction with neighboring communities (small-scale), and GI considers the relationship between the chosen community and all other communities (large-scale). Such comprehensive consideration of connectivity information will enable better measurement of the community vulnerability.

Definition 7: The proposed community vulnerability measurement methodology ζ_i and relative vulnerability ξ_i are defined as follows:

$$\zeta_i = \frac{1}{(\eta_i)^{\alpha}(\sigma_i)^{\beta}(\gamma_i)^{\chi}} \tag{10}$$

$$\xi_i = \frac{\zeta_i}{\min\limits_t \zeta_t} \tag{11}$$

where η_i, σ_i, and γ_i are normalized EIC, EOC, and GI, respectively, allowing information related to the communities to be fully accounted for. The reason for normalizing these factors is to consider them at the same scale. In addition, α, β, and χ are the weighting parameters for different factors.

The setting of weighting parameters is to evaluate the vulnerability in different situations, which can easily adjust the weight based on different information. These weighting parameters make GBCVE more accessible, and some special cases for ζ_i are shown as follows.

1) $\alpha = \beta = \chi$: ζ_i will consider all factors equally.
2) $\beta = 1, \alpha = \chi = 0$: ζ_i will be equivalent to the classical community vulnerability method V_i [63], that is, ζ_i only considers the interaction with neighboring community.
3) $\chi = 0$: ζ_i will consider the information in small scale, including the interior information of community and the interaction with neighboring community.
4) $\alpha = \beta = 0$: ζ_i will consider the information in large scale, that is, the relationship between chosen community and all other communities.

Example 1: Our proposed method ζ_i and classical method V_i are used to evaluate the vulnerability in each community of the social network given in Fig. 1. The weighting factors α, β, and χ equal to 1 in ζ_i will allow us to consider these factors equally. The vulnerability ζ_i and relative vulnerability ξ_i obtained by this GBCVE model, and the result by classical measurement V_x (vulnerability) and R_x (relative vulnerability) are shown in Table I.

The classical measure V_i can explain that c_3 is the most reliable, but it cannot distinguish between the vulnerability of c_1 and c_2 ($V_1 = V_2$). However, c_1 is more vulnerable than c_2, because there are more nodes in c_2, and c_2 is a fully connected community. It is important to highlight that our proposed method ζ_i can clearly distinguish the vulnerability between all communities ($\zeta_1 > \zeta_2 > \zeta_3$). Therefore, our proposed methodology is a more effective method to evaluate the vulnerability of each community.

2) Community Vulnerability Fuzzy Ranking: To better compare the property of each community using vulnerability, it is possible to calculate the fuzzy ranking for each community.

Definition 8: The fuzzy ranking order for community vulnerability is given as follows:

$$\left[c_{f(1)}\ \Omega_1\ c_{f(2)} \cdots c_{f(|C|-1)}\ \Omega_{|C|-1}\ c_{f(|C|)}\right] \tag{12}$$

where $f(\cdot)$: $\{1, 2, \ldots, |C| - 1\} \rightarrow \{1, 2, \ldots, |C| - 1\}$ is a permutation function that ranks community relative vulnerabilities in ascending order, that is, $\xi_{f(t+1)} \geq \xi_{f(t)}$ for any $t \in \{1, 2, \ldots, |C| - 1\}$. $\Omega_1, \ldots, \Omega_{|C|-1}$ is the corresponding relationship between communities, which is the element of $\{\approx, \leq, <, \ll\}$. Inspired by [67], the corresponding relationship transformation approach can be defined as follows:

$$\Omega_t = \begin{cases} \approx & \Delta_t < 0.25\delta \\ \leq & 0.25\delta < \Delta_t < 0.75\delta \\ < & 0.75\delta < \Delta_t < 1.5\delta \\ \ll & 1.5\delta < \Delta_t \end{cases} \tag{13}$$

where Δ_t is the distinction between adjacent vulnerability, that is, $\Delta_t = \xi_{f(t+1)} - \xi_{f(t)}$. In addition, δ is determined by the distinction between adjacent vulnerability values

$$\delta = \frac{1}{|C| - 1} \sum_{t=1}^{|C|-1} \Delta_t = \frac{1}{|C| - 1} \sum_{t=1}^{|C|-1} \left(\xi_{f(t+1)} - \xi_{f(t)}\right). \tag{14}$$

IV. EVALUATIONS

Real-world complex networks are used as test examples to validate and demonstrate the applicability of our proposed community vulnerability evaluation method. The first network is the Manzi network, a telecommunication network. It is an actual network constructed by 52 switching center in Belgium [68], and these centers are connected based on the heuristic algorithm. The second network is the Italian 380kV power transmission grid network [69]. It is a network often used as a benchmark for vulnerability analysis. Every node represents a power source or substation, and the edge indicates a power transmission line. The third example is an E-mail network [70], depicting the E-mail exchanges (and social relationship) between different members in the University Rovira i Virgili. This E-mail network contains more than 1000 users and the vulnerability can be quantified. These three networks are typical test examples used for benchmarking of reliability evaluation. The basic community features pertaining to each network (for example, the number of edges inside community η_i, outside of each community σ_i, GI γ_i, number of nodes in each community n_i) are given in Table II.

A. Case I: Manzi Network

Fig. 4(a) shows the Manzi network with seven communities and modularity $Q = 0.6316$ (strong community structure). Based on division in the community structure, we

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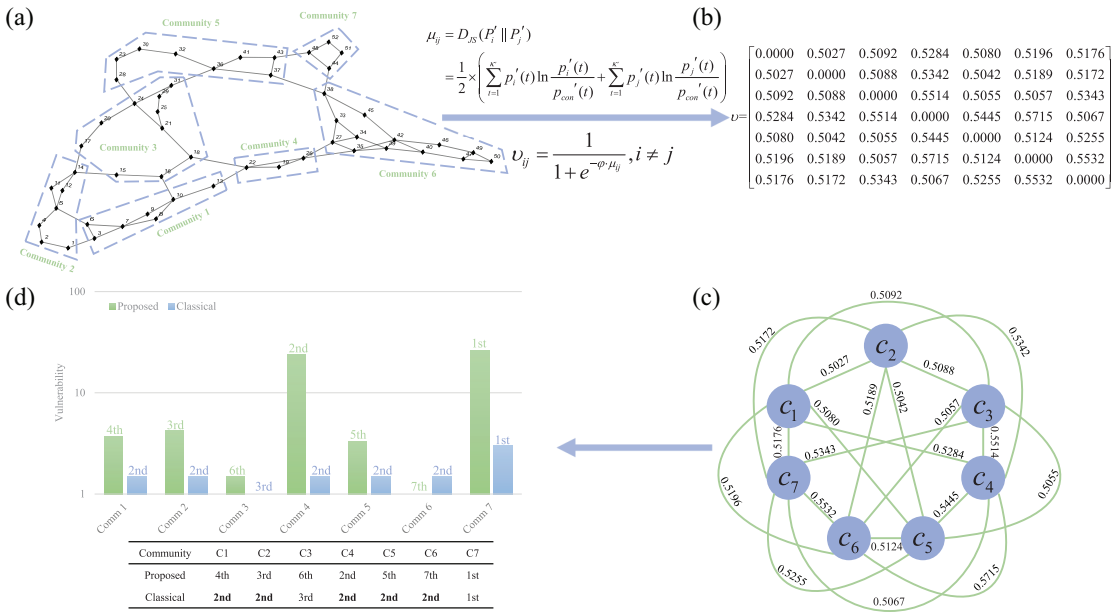

Fig. 4. Properties of Manzi network. (a) Characteristic structure of Manzi network, and the community feature under Newman modularity. (b) Distance obtained by divergence and transition describes the relationship between each pair of communities, and is a symmetric adjacency matrix. (c) Community network is constructed by the adjacency matrix. Each node represents one independent community. (d) Vulnerability result for each community obtained by our proposed method (green) and classical method (blue).

Community	C1	C2	C3	C4	C5	C6	C7
Proposed	4th	3rd	6th	2nd	5th	7th	1st
Classical	**2nd**	**2nd**	3rd	**2nd**	**2nd**	**2nd**	1st

TABLE II
BASIC FEATURES PERTAINING TO THE DIFFERENT NETWORKS

Comunity	Manzi			Italian			Email			
	η_i	σ_i	γ_i	η_i	σ_i	γ_i	η_i	σ_i	γ_i	n_i
c_1	0.4211	0.6667	0.6409	0.5161	0.3750	0.7061	1.0000	1.0000	1.0000	319
c_2	0.3684	0.6667	0.6426	0.4194	0.6250	0.6117	0.7431	0.5687	0.7538	207
c_3	0.5263	1.0000	0.8565	0.4839	0.3750	0.6551	0.1783	0.2475	0.4447	114
c_4	0.1579	0.6667	0.2646	0.6129	0.6250	0.7792	0.5750	0.5020	0.6817	186
c_5	0.4211	0.6667	0.7188	0.5806	1.0000	0.7404	0.5436	0.5313	0.8378	238
c_6	1.0000	0.6667	1.0000	0.4194	0.6250	0.5617	0.0165	0.0525	0.0442	14
c_7	0.2105	0.3333	0.3648	0.2258	0.3750	0.3006	0.0299	0.0606	0.0905	26
c_8	–	–	–	0.3871	0.3750	0.5634	0.0016	0.0020	0.0077	3
c_9	–	–	–	1.0000	0.7500	1.0000	0.0196	0.0323	0.0672	20
c_{10}	–	–	–	0.1935	0.1250	0.3567	0.0055	0.0091	0.0168	6

can measure the difference in distance between each pair of communities. The community network can then be constructed and is shown in Fig. 4(c). In the community network, one independent community is represented as a single node, and it is a fully connected undirected weighted network. The weight shows the difference in property between each pair of communities, and the adjacency matrix is shown in Fig. 4(b). The large-scale community relationship GI γ_i can be obtained via the constructed community network. The number of edges inside community η_i and outside of each community σ_i, as well as the GI γ_i are three important factors considered in our model for evaluating the community vulnerability. These three factors correspond to the interior information of the community, small-scale interaction relationship, and large-scale interaction relationship, respectively. Due to the different scale of η_i, σ_i, and γ_i, they should be normalized first before evaluating for community vulnerability. With equal consideration for all factors (α, β, and χ equal to 1), the vulnerability community results evaluated by our proposed method and classical method [63] are given in Fig. 4(d). Finally, we can obtain the community vulnerability fuzzy ranking order based on the relative vulnerability: $c_6 \approx c_3 \leq c_5 \approx c_1 \approx c_2 \ll c_4 \leq c_7$.

From these comparisons, it is found that c_7 is the most vulnerable community recognized by both classical and our proposed method. c_7 is the most vulnerable community because there is only a ring of four nodes in this community. Next, using the classical method, we observe that c_1, c_2, c_4, c_5, and c_6 [bold form in Fig. 4(d)] tie as the second most vulnerable community, but this cannot be valid due to their entirely different structure and anti-attack ability. On the contrary, our proposed method is able to comprehensively consider the properties of each community and give a specific ranking to each of them. Based on our proposed method, c_4 is the single second most vulnerable (fully connected network with only three nodes). This structure is simpler compared to several other communities, which is supported by the fuzzy ranking result. The reason why c_7 is more vulnerable than c_4 is that c_7 can be entirely replaced by the edge between node 37 and node 38. The vulnerability of c_1, c_2, and c_5 is close (similar with classical method), but our proposed method is still able to give a specific ranking, that is, $\xi_5 < \xi_1 < \xi_2$, which is more reasonable than the classical method that gives equal vulnerability. The vulnerability of c_1, c_2, and c_5 can be observed from the community structure [Fig. 4(a)], such as the number of ring in the community and the external connectivity. Another difference between these two methods is the result pertaining to c_6 and c_3. Our proposed method thinks c_3 is more vulnerable than c_6, but the classical method gives a different ranking. It is worth noting that there is only a ring in c_3, and one important node (Node 14) is located in c_2, which makes numerous nodes connected in a straight line. Therefore, the line structure of c_3 will mean a high probability of being split into two subgraphs after the random node attack. On the contrary, most of nodes in c_6 are connected in a multiring structure, and it will rarely be divided into several subgraphs

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TABLE III
SENSITIVITY ANALYSIS RESULTS OF VULNERABILITY OF COMMUNITIES WITH WEIGHTING PARAMETERS α, β, AND χ

Community	$S_i(\alpha)$	$S_{T_i}(\alpha)$	$S_i(\beta)$	$S_{T_i}(\beta)$	$S_i(\chi)$	$S_{T_i}(\chi)$
c_1	0.4138	0.7251	0.1087	0.3145	0.1263	0.3548
c_2	0.4402	0.7691	0.0921	0.3015	0.1054	0.3378
c_3	0.8996	0.9405	0.0000	0.0000	0.0597	0.1009
c_4	0.1992	0.8234	0.0171	0.2478	0.1232	0.7342
c_5	0.4885	0.7603	0.1283	0.3288	0.0845	0.2381
c_6	0.0000	0.0000	0.9999	0.9998	0.0000	0.0000
c_7	0.1160	0.7588	0.0754	0.6760	0.0582	0.6196

after node attack. Therefore, it can be clearly seen that c_3 is more vulnerable than c_6. Clearly, our proposed method is able to give a specific and realistic ranking result that corresponds with the actual structure property, followed by the fuzzy ranking algorithm to determine the difference between the vulnerability of communities.

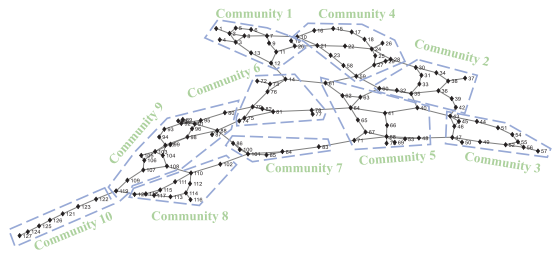
The first-order Sobol' index S_i and total effect index S_{T_i} are then used to analyze the sensitivity of these weighting parameters. As the three weighting parameters can change the vulnerability of communities under different values, these parameters are very crucial for vulnerability measurement. Thus, the strategy to determine these weighting parameters is an important consideration in this model. The first-order Sobol' index S_i and total effect index S_{T_i} used in this article are defined as

$$S_i = \frac{\mathrm{Var}_{X_i}\left(E_{X_{\sim i}}(Y|X_i)\right)}{\mathrm{Var}(Y)}$$
$$S_{T_i} = \frac{E_{X_{\sim i}}\left(\mathrm{Var}_{X_i}(Y|X_{\sim i})\right)}{\mathrm{Var}(Y)} \tag{15}$$

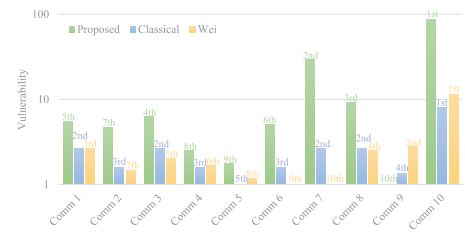
where X_i is the ith independent factor of input X, $X_{\sim i}$ refers to all inputs except X_i, Y is the output of this model, and $\mathrm{Var}(Y)$ is the variance of Y with respect to the change in input. Specifically, S_i can describe the contribution of X_i to Y. S_{T_i} can describe the contribution of the variability of X_i to the variance of Y, that is, $\mathrm{Var}(Y)$. In addition, S_{T_i} considers both individual input effects and the interaction with other inputs.

Different values of weighting parameters will give focus to different aspects of the information related to the community, and thus, give different vulnerability results for networks. The sensitivity analysis results of vulnerability of communities with three weighting parameters are shown in Table III, and some conclusions can be obtained from these results.

1) The value of S_i can show the sensitivity for each weighting parameter. For example, the vulnerability of community c_1 is most sensitive to α, followed by χ, and β is the least sensitive.
2) When the value of factor equals to 1, S_i and S_{T_i} of weighting parameter will be equal to 0, like $\eta_6 = 1$ causes $S_6(\alpha) = 0$, $S_{T_6}(\alpha) = 0$. This means that when the factor is equal to 1, the sensitivity of weighting parameters (index) will not change and is equal to 0.
3) $S_i(\alpha)$ is larger than $S_i(\beta)$ and $S_i(\chi)$, which means this method is more sensitive with α.
4) In most cases, the sum of S_i over three weighting parameters in each community is less than 1. This situation is due to the interaction between these parameters under consideration, but this situation does not appear in S_{T_i}.
5) Even if there is an interaction that makes S_i and S_{T_i} different, it is interesting to note that S_i and S_{T_i} maintain the same order between different information factors in the same community. For example, in c_1, $S_1(\alpha) > S_1(\chi) > S_1(\beta)$ and $S_{T_1}(\alpha) > S_{T_1}(\beta) > S_{T_1}(\chi)$.



(a)



Community	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
Proposed	5th	7th	4th	8th	9th	6th	2nd	3rd	10th	1st
Classical	**2nd**	3rd	**2nd**	3rd	5th	3rd	**2nd**	**2nd**	4th	1st
Wei	3rd	7th	5th	6th	8th	9th	10th	4th	2nd	1st

(b)

Fig. 5. Properties of Italian network. (a) Characteristic structure of Italian network, and the community feature divided by Newman modularity. (b) Vulnerability result obtained by our proposed method (green), classical method (blue), and Wei method (yellow).

B. Case II: Italian Network

Italian power transmission grid network, a commonly used vulnerability evaluation network, is divided into ten communities with modularity $Q = 0.7596$. The structure is shown in Fig. 5(a). Similarly, α, β, and χ are set to 1 so as to consider the three factors in the same scale. The vulnerability of each community obtained by different methods is shown in Fig. 5(b), and we consider one more comparison method, Wei *et al.* [64] method, in this network. The detailed ranking results are shown in Table IV.

It can be seen that c_{10} is the most vulnerable community in the network evaluated by all three methods, because of the line structure of c_{10}. Meanwhile, we can find that the Wei method gives a result that is very different from the other two methods. Specifically, it gives the exact opposite evaluation of c_9 and c_7 as shown in Table IV, so we now attempt to give one detailed comparison between these two communities. We find that c_7 has a very simple structure, that is, a three-node ring and other nodes form a line. However, c_9 has a large number of nodes and a complicated topological structure, with the strong reliability shown by numerous ring nesting structures. Therefore, the Wei method is inadequate here, and the

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TABLE IV
VULNERABILITY RANKING OF COMMUNITIES OBTAINED BY DIFFERENT METHODS AND DIFFERENT WEIGHTING PARAMETERS IN ITALIAN NETWORK

Method	Vulnerability Ranking
Classical method	$R_5 < R_9 < R_2 = R_4 = R_6 < R_1 = R_3 = R_7 = R_8 < R_{10}$
Wei method	$R_7{}' < R_6{}' < R_5{}' < R_2{}' < R_4{}' < R_3{}' < R_8{}' < R_1{}' < R_9{}' < R_{10}{}'$
Proposed method	$\xi_9 < \xi_5 < \xi_4 < \xi_2 < \xi_6 < \xi_1 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Fuzzy ranking of proposed method	$c_9 \approx c_5 \approx c_4 \approx c_2 \approx c_6 \approx c_1 \approx c_3 \leq c_8 \ll c_7 \ll c_{10}$
Special cases	**Vulnerability Ranking**
Focus on large-scale information ($\alpha = 1, \beta = 1, \chi = 2$)	$\xi_9 < \xi_5 < \xi_4 < \xi_2 < \xi_1 < \xi_6 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Focus on small-scale information ($\alpha = 1, \beta = 2, \chi = 1$)	$\xi_9 < \xi_5 < \xi_4 < \xi_2 < \xi_6 < \xi_1 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Focus on interior information ($\alpha = 2, \beta = 1, \chi = 1$)	$\xi_9 < \xi_5 < \xi_4 < \xi_1 < \xi_2 < \xi_6 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Ablation analysis	**Vulnerability Ranking**
Omit large-scale information ($\alpha = 1, \beta = 1, \chi = 0$)	$\xi_9 < \xi_5 < \xi_4 < \xi_2 < \xi_6 < \xi_1 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Omit small-scale information ($\alpha = 1, \beta = 0, \chi = 1$)	$\xi_9 < \xi_4 < \xi_5 < \xi_1 < \xi_3 < \xi_2 < \xi_6 < \xi_8 < \xi_{10} < \xi_7$
Omit interior information ($\alpha = 0, \beta = 1, \chi = 1$)	$\xi_9 < \xi_5 < \xi_4 < \xi_2 < \xi_6 < \xi_1 < \xi_3 < \xi_8 < \xi_7 < \xi_{10}$
Only consider large-scale information ($\alpha = 0, \beta = 0, \chi = 1$)	$\xi_9 < \xi_4 < \xi_5 < \xi_1 < \xi_3 < \xi_2 < \xi_8 < \xi_6 < \xi_{10} < \xi_7$
Only consider small-scale information ($\alpha = 0, \beta = 1, \chi = 0$)	$\xi_5 < \xi_9 < \xi_2 = \xi_4 = \xi_6 < \xi_1 = \xi_3 = \xi_7 = \xi_8 < \xi_{10}$
Only consider interior information ($\alpha = 1, \beta = 0, \chi = 0$)	$\xi_9 < \xi_4 < \xi_5 < \xi_1 < \xi_3 < \xi_2 < \xi_6 < \xi_8 < \xi_7 < \xi_{10}$

rest of the ranking by this method will not proceed. We can find that the classical method gives equal vulnerability of c_1, c_3, c_7, and c_8 [bold form in Fig. 5(b)], which cannot distinguish the difference between their vulnerability. However, our proposed method is able to give a specific ranking of their vulnerability, that is, $\xi_1 < \xi_3 < \xi_8 < \xi_7$, and their fuzzy ranking is: $c_1 \approx c_3 \leq c_8 \ll c_7$. The fuzzy ranking relationship $\ll, \leq, \approx$ gives a specific description of the vulnerability between communities. More details about their vulnerability comparison can be given. c_7 can be easily divided into two subgraphs of approximate size, and disconnected with c_8 due to the existence of node 101. Hence, c_7 will $\gg$ other communities. c_8 is more reliable than c_7 due to the two-ring and balance structure, that is, there is no vulnerable node that will render the community to be easily divided into several subgraphs and disconnect with others. However, c_8 is not reliable enough compared to c_3 and c_1. This is because c_3 and c_1 have more ring nesting structures to ensure the connectivity of each node, thereby avoiding being easily divided. This can be easily observed from the community structure. Meanwhile, due to the similar structure between c_3 and c_1, their characteristic will be approximately the same. Our method can effectively distinguish between the vulnerability of communities, which has not been able to be identified by the classical method. Our method also gives a reasonable ranking and is especially suitable for other communities whose vulnerability cannot be evaluated by classical method as well, that is, c_2, c_4, and c_6 [underline form in Fig. 5(b)]. Compared with the classical method and Wei method, our proposed model is able to give a more effective vulnerability result, which follows closely the rule of community structure, and is able to distinguish communities with similar vulnerabilities.

We now discuss some special cases in Table IV. When our proposed model focuses on the large-scale information, the only difference from the result under equal weighting parameters is the order of c_6 and c_1, and our proposed method under this weighting parameters setting is superior to the classical method and Wei method. From the fuzzy ranking, the vulnerability of c_6 and c_1 is $\approx$, which means their vulnerability is very close. When $\alpha = 2, \beta = 1, \chi = 1$, our proposed method focuses on the interior information. The difference from the result under equal weighting parameters is the order of c_1, c_2, and c_6. It is similar to the large-scale information consideration, and the relationships among these communities are $\approx$ in fuzzy ranking. From the overall ranking, the result in this special case is still better than the other comparison methods. For the small-scale information consideration, the result is entirely the same as the result under equal weighting parameters, which means our proposed method can give reasonable evaluation under different weighting parameters. The ablation study is further conducted to analyze the importance of each factor. When omitting the large-scale information ($\alpha = 1, \beta = 1$, and $\chi = 0$) or the interior information ($\alpha = 0, \beta = 1$, and $\chi = 1$), the results are the same as the result obtained by considering all the information. However, omitting the small-scale information will lead to a large difference, especially c_7 is identified as the most vulnerable community, which is the opposite of other results that think c_{10} is the most vulnerable community. When only considering a single factor, many communities will be misidentified. For example, c_6 is more vulnerable than c_1 and c_3 when only considering large-scale information or interior information, which is different from the result obtained by the classical method, Wei method, and our proposed method. In addition, our proposed approach will degenerate to the classical method when only considering the small-scale information ($\alpha = 0, \beta = 1$, and $\chi = 0$) and give many communities with the equal vulnerability. Therefore, each part of information is essential to evaluate the vulnerability of communities.

C. Case III: E-Mail Network

Due to the small size of the first two networks, we have given a more qualitative analysis based on the community structure. The E-mail network to be discussed in this section contains more than 1000 nodes and 5000 edges. We will approach this test example using the quantitative analysis of community vulnerability. The network can be divided into ten communities with the modularity $Q = 0.5037$, and the number of nodes in each community is shown in Table II to express the complexity for each community. The vulnerability results evaluated by the different methods are shown in Fig. 6(a). We can find that almost all ranks are the same except for c_6 and c_9 [bold form in Fig. 6(a)], so we first analyze the reason why they have different ranks. The structure of internal nodes and the directly connected external nodes of these two communities are shown in Fig. 6(c). First, the size of the two communities is small, and there are more nodes in c_9 (20 nodes) than nodes in c_6 (14 nodes) which can thus form more structure in the community. As for the smallest closed shape in each community, we can find that c_6 has six triangles and one polygon, but c_9 has four triangles and three polygons. There are more polygons in c_9 although they have the same number of smallest closed shapes. Therefore, from the structure composition, we can conclude that c_6 is more vulnerable than c_9, thereby demonstrating the superiority of our proposed method. We then give a quantitative analysis of the vulnerability in the E-mail network. We can find that c_1 has a large number

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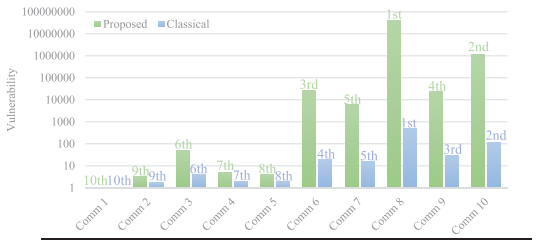
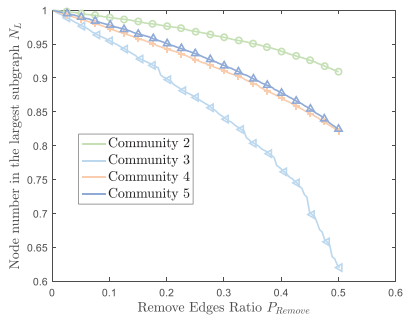
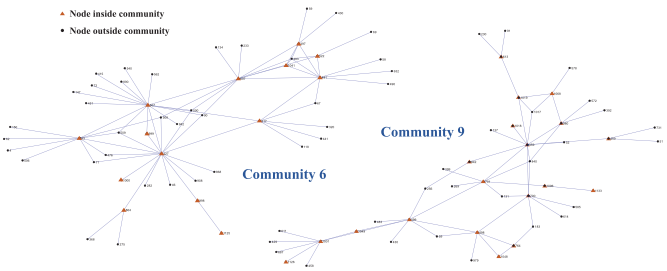


Community	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
Proposed	10st	9th	6th	7th	8th	**3rd**	5th	1st	**4th**	2nd
Classical	10th	9th	6th	7th	8th	**4th**	5th	1st	**3rd**	2nd

(a)



(b)



(c)

Fig. 6. Properties of E-mail network. (a) Vulnerability result of each community obtained by our proposed method (green) and classical method (blue). (b) Number of nodes in the largest subgraph with different ratios (removed nodes over total edges), which is the quantitative analysis of community vulnerability. (c) Topological structure of c_6 and c_9, which contains the internal nodes and the directly connected external nodes with neighbor communities.

of nodes compared to other communities, and the number of nodes in c_6 to c_{10} is very small, which is not conducive for any quantitative analysis. As an illustration, we use c_2 to c_5 for quantitative analysis due to their complicated structure properties. To represent the vulnerability changes of community in quantity, the number of nodes in the largest subgraph [55] N_L is used

$$N_L(P_t) = \frac{\max(N_i(P_t))}{N}, i = s_1, s_2, \ldots, s_n \tag{16}$$

where $s_1, s_2, \ldots, s_n$ are subgraphs, which are divided due to the edge removal, P_t is the ratio of removed edges and total edges in each step; N_L represents the number of nodes in the largest subgraph in each step, obtained by numerous repeated experiments. We set the maximum number of deleted edges to 50% with 100 repetitions. The results are shown in Fig. 6(b). It can be found that c_3 has the fastest decline rate of N_L, far faster than any other communities, and the slowest decline rate is given in c_2. c_4 and c_5 have similar decline rate, but c_4 is faster. The decline rate of N_L can represent the vulnerability of each community and the result is the same as our proposed method. Therefore, from the quantitative analysis, our proposed method gives a similar result with the real-world community vulnerability changes.

TABLE V
RUNNING TIME (IN SECONDS) FOR THREE METHODS IN DIFFERENT NETWORKS

Method	Manzi	Italian	Email
Classical method	4.07×10^{-5}	6.87×10^{-5}	1.21×10^{-3}
Wei method	1.45×10^{-4}	3.08×10^{-4}	8.99×10^{-3}
Proposed method	9.92×10^{-5}	1.86×10^{-4}	1.73×10^{-3}

D. Time Complexity

We now analyze the time complexity of the proposed method across different real-world networks, and the running time is shown in Table V. The time elapsed is obtained from the average of 1000 independent repeated experiments. As the community structure is the objective here, and the time for community detection is the same for all methods, it is reasonable not to consider the running time for community detection in all the methods. Our experiments are obtained using MATLAB 2016a on a PC equipped with an Intel Core i7-10710U CPU operating at 1.61 GHz and 16-GB RAM. As observed from the results, the classical method gives the lowest running time, our proposed method comes in second, and the longest running time is the Wei method. In general, the classical method and our proposed method have similar time complexity. The difference between these two methods becomes smaller as the network size increases, and the time taken is 1.21×10^{-3} and 1.73×10^{-3}, respectively, in the E-mail network. Meanwhile, the Wei method has the longest running time as we can observe from these three networks, and the running time is at least 1.5 times of our proposed method (and even reaching up to five times in the E-mail network). This reveals that our proposed method is well suited and can be implemented on real-world situations. Moreover, the vulnerability result obtained by our proposed method is much more reasonable, and also closely related to the community structure.

V. CONCLUSION

The increase in social media users and widespread community structure makes community vulnerability evaluation an important topic to be investigated. Due to the critical gaps in existing methods (including inadequate information consideration, application limitations, and time complexity), we have proposed an original and novel community vulnerability evaluation model in this article. Our approach is different from existing methods as we considered three important factors: 1) the number of EIC; 2) the number of edges connected to neighboring communities; and 3) the GI. These three factors correspond to the interior information of the community, small-scale interaction relationship, and large-scale interaction relationship, respectively. The first two factors are the basic

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property of the network, which can be easy obtained from the topological structure, and the last factor (large-scale interaction relationship) is the focus of our proposed GBCVE model. The structure difference between each pair of communities is first measured by the Jensen–Shannon divergence. The difference is then transformed to distance by the log-sigmoid transition function ($\varphi = 3$). The GI of each community can be obtained by the gravity model, which reveals the relationship between the chosen community and all other communities. Furthermore, the vulnerability degree of each community can be evaluated by this proposed GBCVE model (weighting parameters equal 1 in general), and the specific vulnerability ranking is given via fuzzy ranking technique. In some cases, our proposed method is equivalent to the classical evaluation method with special setting of weighting parameters. To make our model complete, the global sensitivity of weighting parameters is analyzed by the Sobol' index.

Three real-world complex networks have been used in Section IV as test-examples to evaluate our proposed method. For the Manzi network, there are seven communities to evaluate their vulnerability. We have shown that our proposed model can better evaluate the vulnerability of each community, because it can give different values of ξ_i when the classical method assesses these communities to be of the same vulnerability. In addition, the fuzzy ranking of communities can determine the difference in vulnerability between them. The sensitivity analysis shows which factor is more influential in terms of the vulnerability result. For the Italian network, we have compared our results by the Wei method. The vulnerability result based on our proposed model is more reasonable than these classical methods. For the E-mail network, the number of nodes in the largest subgraph is used as a comparison between these large communities. The comparison results support the rank obtained by our proposed method over the classical methods. The experimental results obtained via the three real-world network test-examples have thus demonstrated the applicability of our proposed community vulnerability evaluation method, and its superiority over existing methods. An important contribution from our work here is that the model can first be applied to various background networks to analyze their basic property, that is, the community vulnerability, before any in-depth research on the context of the problem is being carried out.

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